SATGURU STUDY CENTRE

Chennai - 81

TN CHENNAI DISTRICT STATEBOARD QUESTION PAPER 2024 – 2025

Follow For More Updates

SATGURU STUDY CENTRE

NEVER STOP LEARNING.NEVER STOP GROWING...

CLASS : 10

Register Number | | | | | 3 | 5

FIRST REVISION EXAMINATION, JANUARY - 2025

MATHEMATICS

Time Allowed : 3.00 Hours] [Max. Marks : 100

PART – I

I. Answer all of the following: **14x1=14**

1. A= {a,b,p} , B= {2,3} , C = {p,q,r,s} then n[(A∪C) x B] is
   a) 8 b) 20 c) 12 d) 16
2. If the ordered pairs (a+2, 4) and (5, 2a+b) are equal then (a, b) is
   a) (2, -2) b) (5, 1) c) (2, 3) d) (3, -2)
3. If $f(x) = 2x^2$ and $g(x) = {}^1/_{3x}$ then fog is
   a) $\frac{3}{2x^2}$ b) $\frac{2}{3x^2}$ c) $\frac{2}{9x^2}$ d) $\frac{1}{6x^2}$
4. The least number that is divisible by all the numbers from 1 to 10 is
   a) 2025 b) 5220 c) 5025 d) 2520
5. $7^{4k} \equiv$ ---------------- (mod 100)
   a) 1 b) 2 c) 3 d) 4
6. The value of $(1^3+2^3+3^3+\ldots\ldots+15^3) - (1+2+3+\ldots\ldots+15)$ is
   a) 14400 b) 14200 c) 14280 d) 14520
7. The solution of the system $x+y-3z=-6$, $-7y+7z=7$, $3z=9$ is
   a) $x=1$, $y=2$, $z=3$ b) $x=-1$, $y=2$, $z=3$ c) $x=-1$, $y=-2$, $z=3$ d) $x=1$, $y=-2$, $z=3$
8. The solution of $(2x-1)^2 = 9$ is equal to
   a) -1 b) 2 c) -1, 2 d) None of these
9. In ΔLMN, ∠L=60°, ∠M=50°. If ΔLMN ~ ΔPQR then the value of ∠R is
   a) 40° b) 70° c) 30° d) 110°
10. If in ΔABC , DE||BC, AB=3.6cm, AC=2.4cm and AD=2.1cm then the length of AE is
    a) 1.4cm b) 1.8cm c) 1.2cm d) 1.05cm
11. If (5, 7) , (3, P) and (6, 6) are Collinear, then the value of P is
    a) 3 b) 6 c) 9 d) 12
12. The slope of the line which is Perpendicular to a line joining the points (0, 0) and (-8, 8) is
    a) -1 b) 1 c) ${}^1/_3$ d) -8
13. $\tan\theta \operatorname{cosec}^2\theta - \tan\theta$ is equal to
    a) $\sec\theta$ b) $\cot^2\theta$ c) $\sin\theta$ d) $\cot\theta$
14. If $\sin\theta = \cos\theta$, then $2\tan^2\theta + \sin^2\theta - 1$ is equal to
    a) ${}^{-3}/_2$ b) ${}^3/_2$ c) ${}^2/_3$ d) ${}^{-2}/_3$

PART - II

II. Answer any 10 questions. Question No. 28 is compulsory. **10x2=20**

15. If AxB = {(3, 2), (3, 4), (5, 2), (5, 4)} then find A and B.
16. Let $f(x) = 2x+5$. If $x \neq 0$ then find $\frac{f(x+2) - f(2)}{x}$
17. If $13824 = 2^a \times 3^b$ then find a and b.
18. Find the sum of infinity of 9+3+1+..........
19. Find the excluded value of v $\frac{7p + 2}{8p^2+13p+5}$
20. Simplify : $\frac{5t^3}{4t - 8} \times \frac{6t - 12}{10t}$
21. Determine the nature of the roots for $\sqrt{2}\,t^2 - 3t + 3\sqrt{2} = 0$.

22. In ΔABC, if $DE \| BC$, $AD = x$, $DB = x-2$, $AE = x+2$ and $EC = x-1$ then find the lengths of the sides AB and AC.
23. Show that the points P(-1.5, 3), Q(6, -2), R(-3, 4) are Collinear.
24. What is the inclination of a line whose slope is 1.
25. A Cat is located at the point (-6, -4) in xy plane. A bottle of milk is kept at (5, 11). The Cat wish to consume the milk travelling through shortest possible distance. Find the equation of the path it needs take its milk.
26. Prove that $\sec\theta - \cos\theta = \tan\theta \sin\theta$.
27. Show that $\sqrt{\dfrac{1+\sin\theta}{1-\sin\theta}} = \sec\theta + \tan\theta$
28. When the positive integers a, b and c are divided by 13 the respective remainders are 9, 7 and 10. Find the remainder when a+2b+3c is devided by 13.

PART - III

III. Answer the following any 10 questions. Q.No.42 is compulsory. **10x5=50**

29. Let $A = \{x \in W \mid x < 2\}$, $B = \{x \in N \mid 1 < x \leq 4\}$ and $C = \{3, 5\}$ verify that $A\times(B\cap C) = (A\times B)\cap(A\times C)$.
30. If the function $f : R \to R$ is defined by $f(x) = \begin{cases} 2x+7; & x<-2 \\ x^2-2; & -2 \leq x < 3 \\ 3x-2; & x \geq 3 \end{cases}$ then find the values by

    i) f(4) ii) f(-2) iii) f(4) + 2f(1) iv) $\dfrac{f(1) - 3f(4)}{f(-3)}$
31. If $f(x) = x-1$, $g(x) = 3x+1$ and $h(x) = x^2$ then show that $(f\circ g)\circ h = f\circ(g\circ h)$.
32. Find the HCF of 396, 504, 636.
33. In an AP sum of four consecutive terms is 28 and the sum of their squares is 276. Find the four numbres.
34. If a, b, c are three consective terms of an A.P and x, y, z are consecutive terms of a GP then prove that $x^{b-c} \times y^{c-a} \times z^{a-b} = 1$.
35. Find the GCD of the Polynomials x^3+x^2-x+2 and $2x^3-5x^2+5x-3$.
36. If $4x^4-12x^3+37x^2+bx+a$ is a perfect square then find the value of a and b.
37. If the difference between a number and its reciprocal is $^{24}/_{5}$, find the number.
38. Find the value of k, if the area of a quadrilateral is 28 sq.units, Whose vertices are taken in the order (-4, -2), (-3, k), (3, -2) and (2, 3).
39. Without using Pythagoras theorem, show that the points (1, -4), (2, -3) and (4, -7) form a right angled triangle.
40. Find the equation of the Perpendicular bisector of the line joining the points A(-4, 2) and B(6, -4).
41. If $\operatorname{cosec}\theta + \cot\theta = P$ then prove that $\cos\theta = \dfrac{P^2-1}{P^2+1}$
42. Two poles of height 'a' metres and 'b' metres are 'p' metres apart. Prove that the height of the point of intersection of the lines joining the top of each pole to the foot of the opposite pole is given by $\dfrac{ab}{a+b}$ meters

PART - IV

IV. Answer the following. **2x8=16**

43. a) Construct a triangle simillar to a given triangle PQR with its sides equal to $^{7}/_{4}$ of the corresponding sides of the triangle PQR (Scale factor $^{7}/_{4} > 1$) **(OR)**

    b) Construct a ΔPQR with the base PQ=4.5cm, $\angle R = 35^\circ$ and the median RG from R to PQ is 6cm.
44. a) A bus is travelling at a function speed of 50km/hr. Draw the distance-time graph and hence find

    i) The constant of variation.

    ii) How far will it travel in 90 minutes?

    iii) The time required to cover a distance of 300km from the graph. **(OR)**

    b) Graph the following linear equation $y = {}^{1}/_{2}x$. Identify the constant of variation and verify it with the graph. Also i) Find y when x = 9, ii) Find x when y = 7.5